

CSE-3103: Microprocessor and Microcontroller

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8259A Programmable Interrupt Controller

Status Register →

three 8-bit status registers are readable in 8259A.

(1) Interrupt request register →

indicates which interrupt request inputs are active.

(2) In-service register →

contains level of interrupt being serviced.

(3) Interrupt mask register →

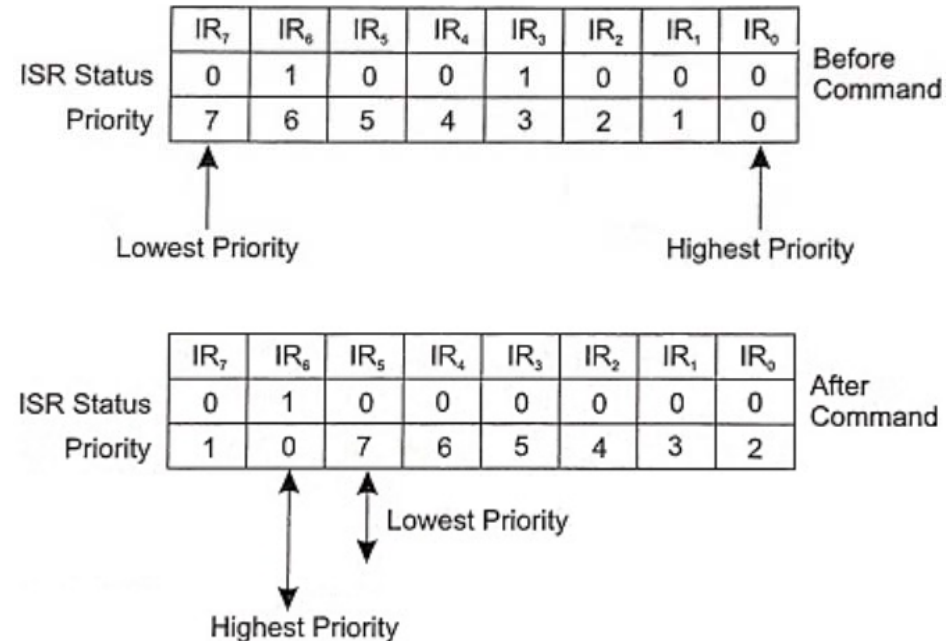
indicates which interrupts are masked off.

IRR and ISR are read by programming OCW_3 ;

$A_0 = 0$; D_0 and D_1 select which register is read.

IMR is read through OCW_1 ;

$A_0 = 1$.



Intel 80186 Microprocessors

Intel 80186 and 80286 →

enhanced versions of 80x86 family of microprocessors.

16-bit microprocessors,

upward-compatible with 8086.

hardware is similar to earlier versions.

Intel 80186 →

16-bit data bus, 20-bit address bus.

internal register structure = 8086.

contain → additional reserved interrupt vectors,

very powerful built-in I/O features.

= embedded controllers.

versions →

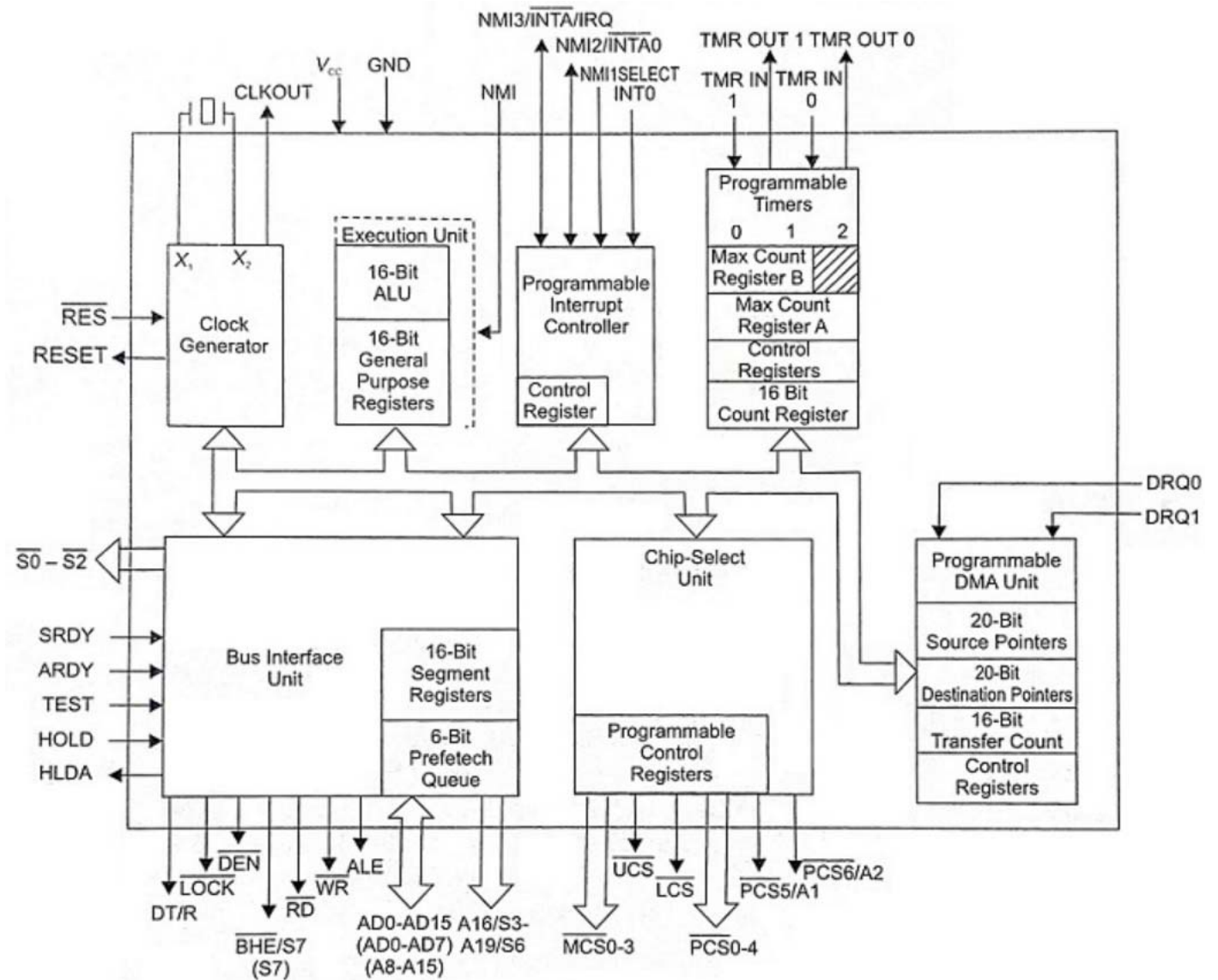
80C186XL: extended temp range and enhanced speed,

80C186EA: enhanced architecture,

80C186EB: further refined,

80C186EC: enhanced, complete.

Basic block diagram
of 80186 →



Intel 80186 Microprocessors

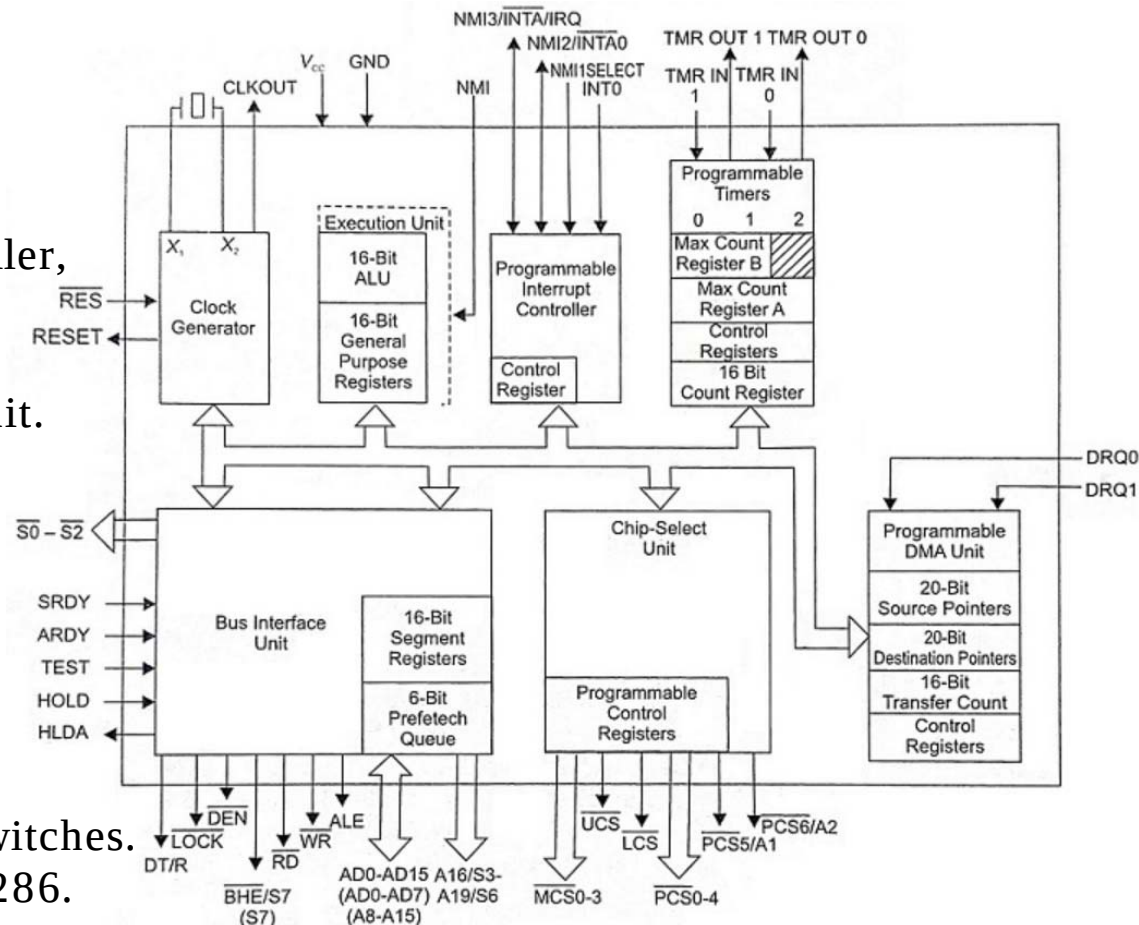
Prefetch queue = 6 bytes.

Contains →

BIU and EU,
clock generator,
programmable interrupt controller,
programmable timers,
programmable DMA controller,
programmable chip selection unit.

Enhancements benefits →

greatly increases system utility,
reduces number of
peripheral components.
caching disk controllers,
LAN controllers.
cellular telephone network as switches.
software for 80186 = that of 80286.



Intel 80186 Microprocessors

Basic features →

Clock Generator →

internal clock generator replaces external 8284A clock generator.

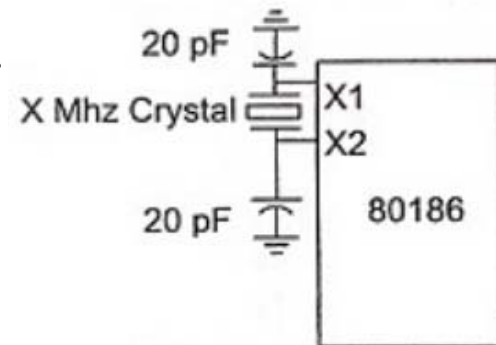
clock pins are connected to crystal →

twice operating frequency of microprocessor.

CLKOUT pin →

system clock signal = $\frac{1}{2}$ of crystal frequency,

duty cycle = 50%.



Programmable Interrupt Controller (PIC) →

arbitrates internal and external interrupts,

controls up to 2 external 8259A PICs →

external 8259 = slave.

internal PIC = master.

without external 8259 →

5 interrupt inputs: INT0 to INT3, NMI.

Intel 80186 Microprocessors

Basic features →

Timers →

contains 3 fully programmable 16-bit timers.

timers 0 and 1 →

generate waveforms for external use,
driven by either master clock of 80186 or
by external clock.

used to count external events.

timer 2 →

internal and clocked by master clock.

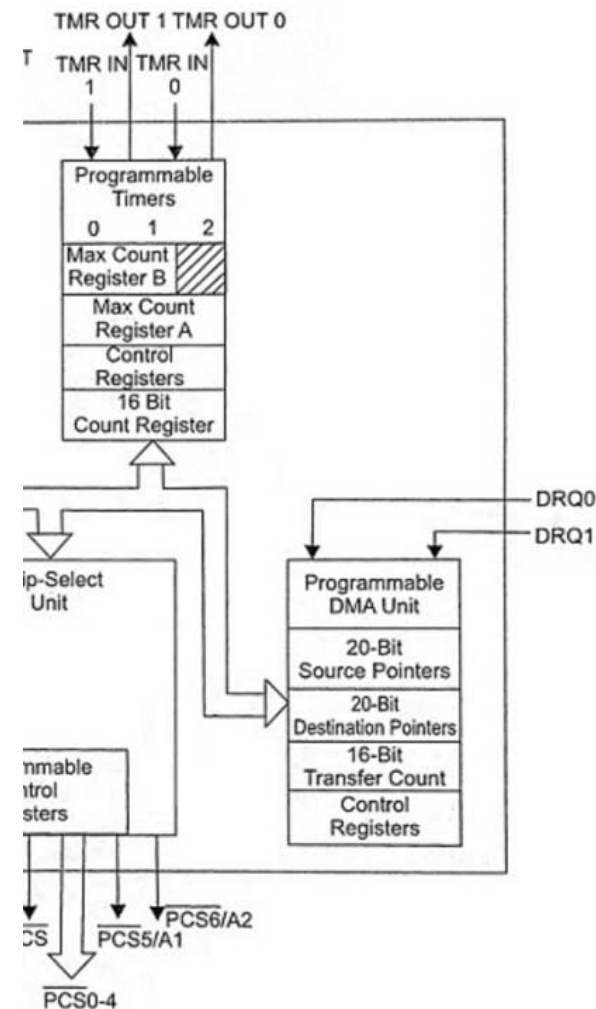
generates interrupt after specified number of clocks,
can provide clock to other timers.

Programmable DMA Unit →

contains 2 DMA channels.

each channel can transfer data between →

memory locations,
memory and I/O,
I/O devices.



Intel 80186 Microprocessors

Basic features →

Programmable Chip Selection Unit →

= built-in programmable memory and I/O address decoder.

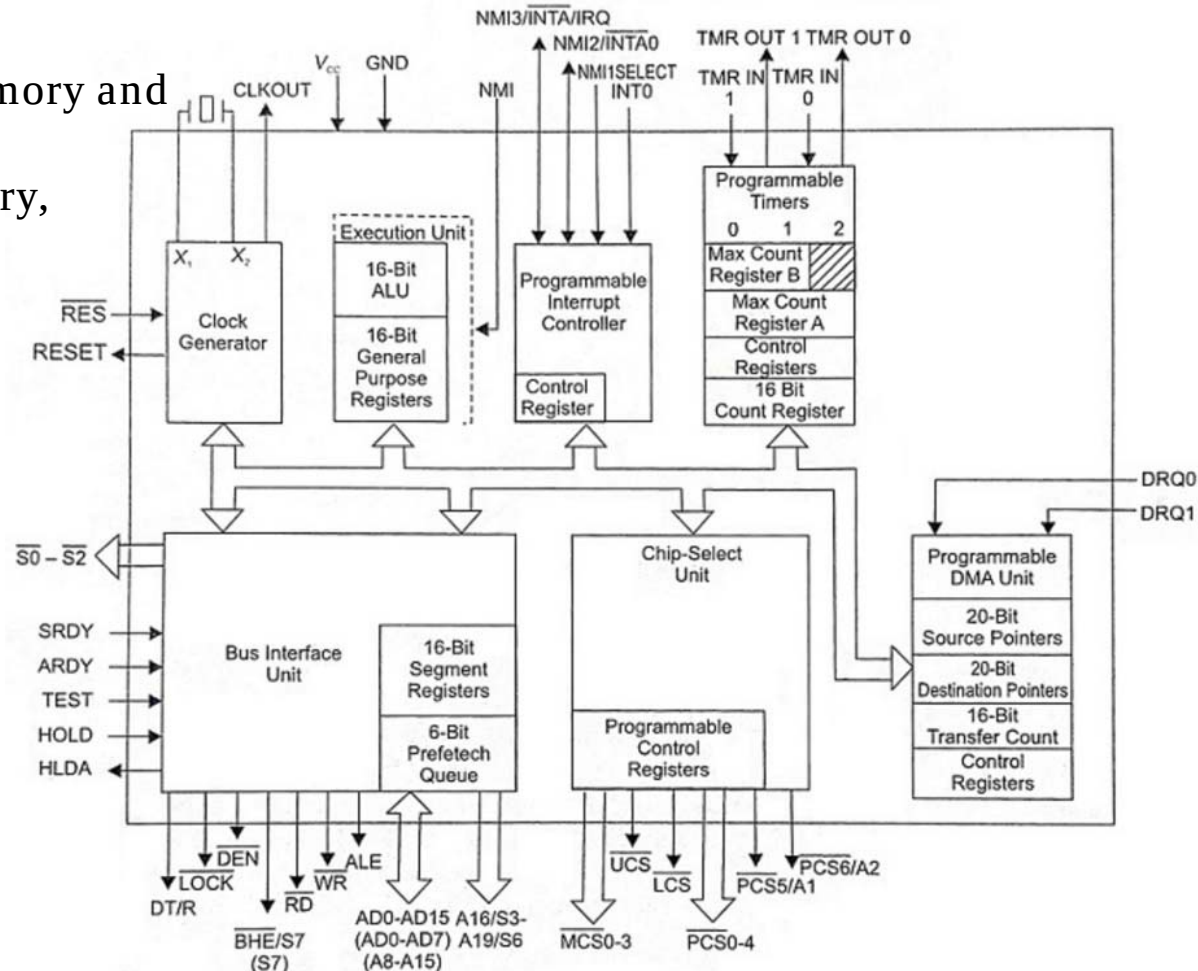
6 output lines to select memory,
7 lines to select I/O.

Power Save Feature →

allows system clock to be divided by 4, 8, or 16,
reduce power consumption.
started by software,
exited by interrupt.

Power Down Feature →

stops clock completely,
entered by execution of
HLT instruction,
exited by any interrupt.



Intel 80186 Microprocessors

Basic features →

Refresh Control Unit →

- automatically refreshes dynamic RAM (DRAM).

- generates refresh row address at interval programmed.

- refresh address + control signal is provided to memory system.

- memory system runs refresh cycle during active time of control signal.

Peripheral Control Block (PCB) →

- set of 256 registers located in I/O or memory space.

- 16-bit-wide registers control all internal peripherals.

80186 is reset →

- PCB is automatically located at top of I/O map,

- I/O addresses = FF00H–FFFFH.

PCB may be relocated at any time to any other area of memory or I/O.

relocation →

- changing contents of relocation register,

- offset addresses = FEH and FFH.

Intel 80186 Microprocessors

Basic features →

Interrupts in 80186 →

interrupts in 80186 = interrupts in 8086,
additional interrupt vectors →

some of internal devices.

array BOUND, unused opcode, ESC opcode, ... etc.

Intel 80286 Microprocessors

Features →

- = iAPX 286 (Intel Advanced Performance Architecture),

- x86 16-bit microprocessor with 134,000 transistors.

- clock frequency = 6 MHz (0.9 MIPS), 8 MHz (1.5 MIPS), 12.5 MHz (1.8 MIPS).

- 16-bit data bus, 16-bit internal register.

- 24-bit address bus →

 - able to address up to 16 MB of memory.

- can run →

 - multitasking applications,

 - digital communications,

 - real-time process control systems,

 - multi-user systems.

- no on-chip clock generator circuit.

- external 82284 chip →

 - generates external clock.

 - clock is divided by 2 internally to generate internal clock.

 - provides 80286 RESET and READY signals.

Intel 80286 Microprocessors

Features →

- 1st x86 processor to operate in protected mode.

- operates in two different modes →

 - real mode and protected mode.

- real mode →

 - compatibility with existing 8086 software base,

 - 80286 is booted in real mode,

 - not possible to switch it from protected mode to real mode.

- protected mode →

 - enhanced system level features.

 - memory management: 24-bit address, access up to 16 MB memory,

 - multitasking: run multiple programs safely,

 - protection: protect each program from interfering with others.

Intel 80386 Microprocessors

Features →

- manufactured by Intel using 0.8-micron CHMOS technology.

- 32-bit microprocessor =

 - 32-bit internal and external data bus,

 - 32-bit registers.

- support 8-bit, 16-bit, 32-bit operands.

- 32-bit address bus →

 - can address up to 4 GB of physical memory.

- physical memory →

 - organized in segments,

 - segment size = 4 GB maximum.

 - support 16k number of segments,

 - total virtual memory space = $4 \text{ GB} \times 16\text{k} = 64 \text{ TB}$.

- 16-byte prefetch queue.

- clock speeds = 16 MHz to 33 MHz.

- operates in →

 - real, protected and virtual real mode.

Intel 80386 Microprocessors

Features →

memory management unit (MMU) =
responsible for handling memory addresses.

1) segmentation unit →

gives protection to data or code present in memory.

provides 4 privilege levels (or protection rings) →

highest: operating system kernel,

high: device drivers,

medium: system utilities,

lowest: user applications

2) paging unit →

operates only in protected mode,

changes linear address into physical address.

provides virtual memory and memory isolation.

instruction set is upward compatible.

supports Intel 80387 numeric data processor.